Assignment3 – Decisions

Given: May 6 Due: May 11

Problem 3.1 (Bayesian Networks in Python)

The goal of this exercise is to implement inference by enumeration in Bayesian networks in Python. You can find the necessary files at `https://kwarc.info/teaching/AI/resources/AI2/bayes/`.

Your task is to implement the `query` function in `bayes.py`. Use `test.py` for testing your implementation.

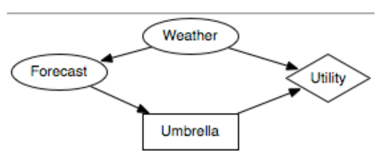
Hint: First implement a function for the full joint probability distribution.

Problem 3.2 (Decision Network)

You try to decide on whether to take an umbrella (Boolean variable M) to Uni. Obviously, it is useful (numeric variable U for the utility) to do so if it rains (Boolean variable W) when you go back home, but it is annoying to carry around if it does not even rain. You decide based on whether the weather forecast predicts rain (Boolean variable F).

1. Draw the decision network for bringing/leaving an umbrella depending on the weather forecast and the actual weather. Explain the four variables M, U, W, and F and what to store in their probability tables.

Solution:



We use four random variables:

- Boolean F for the weather forecast (true if rainy).
- Boolean W for the weather (true if rainy).
- Number U for the utility. U is a utility node and as such is a deterministic variable. Instead of a probability table $P(U = u \mid W = w, M = m)$, we store the function $U(w, m)$ that yields the utility u of (not) having an umbrella if it is (not) rainy.
- Boolean M for whether we took an umbrella. As a Bayesian network variable, M is influenced by F. But because it is a decision variable, there is no probability table $P(M \mid F)$ for it. Instead, the value of M is a decision that we make based on the value of F.

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2. Explain formally how to compute whether or not to take an umbrella, assuming you know the probability that the forecast is correct.

Solution: The probability of the forecast being correct gives us $P(W = b \mid F = b)$ for booleans b. Now given evidence $F = b$, we plug in each possible value for the decision variable M and compute the value of the utility variable.

- $M = true$

$$P(W = true \mid F = b)U(W = true, M = true) +$$

$$P(W = false \mid F = b)U(W = false, M = true)$$

- $M = false$

$$P(W = true \mid F = b)U(W = true, M = false) +$$

$$P(W = false \mid F = b)U(W = false, M = false)$$

We choose the decision that yields higher utility. Note that we may arrive at different decisions for different values of b.

Problem 3.3 (Decision Preferences)

1. Name and state three of the axioms for preferences (i.e. $\succ$).

Solution: Pick any three of the following:

Orderability: $A \succ B \vee B \succ A \vee A \sim B$

Transitivity: $A \succ B \wedge B \succ C \Rightarrow A \succ C$

Continuity: $A \succ B \succ C \Rightarrow (\exists p.[p,A;1-p,C] \sim B)$

Substitutability: $A \sim B \Rightarrow [p,A;1-p,C] \sim [p,B;1-p,C]$

Monotonicity: $A \succ B \Rightarrow ((p > q) \Leftrightarrow [p,A;1-p,B] \succeq [q,A;1-q,B])$

Decomposability: $[p,A;1-p,[q,B;1-q,C]] \sim [p,A\,;((1-p)q),B\,;((1-p)(1-q)),C]$

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2. How are preferences related to value functions?

Solution: Ramsey's theorem states that given a set of preferences that obey the constraints above, there is a value function U with

$$(U(A) \geq U(B)) \Leftrightarrow A \succeq B \quad \text{and} \quad U([p_1,S_1 ; \ldots ; p_n,S_n]) = \sum_i p_i U(S_i)$$

Problem 3.4 (Decision Network)

You need a new car. Your local dealership has two models on offer

- C_1 for \$1500 with market value \$2000
- C_2 for \$1150 with market value \$1400

Either car can be of good quality or bad quality, and you have no information about that. If C_1 is of bad quality, repairing it will cost \$700, if C_2 is of bad quality repairing it will cost \$150.

You have the choice between two tests:

1. $Test_1$ at cost \$50: This will confirm that C_1 is of good quality (if it is) with certainty 85% probability, and that it is of bad quality (if it is) with certainty 65%.

2. $Test_2$ at cost \$20: This will confirm that C_2 is of good quality (if it is) with 75% certainty, and that it is of bad quality (if it is) with certainty 70%.

The a priori probability (without any tests) that a car is of good quality is 70% for C_1 and 65% for C_2.

The utility function is the monetary value, i.e., the difference of the market value of the acquired car and the amount of money spent on test, car, and repair.

1. Decision networks in general have three kinds of nodes. Explain the differences regarding the probability tables of the three kinds.

Solution: Decision nodes have no probability table. Instead, we fix the value for each decision, then calculate the utility. Utility nodes are deterministic: their value can be computed if the values of their parent nodes are known. In particular, we can compute their expected value if the probabilities for their parent nodes are known. Chance nodes have the usual probability tables of Bayesian networks.

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2. Now regarding the concrete network used here, explain the random variables of all nodes and their domains.

Solution: The variables are

- $C_1, C_2 \in \{g, b\}$: car 1 resp. car 2 are of good/bad quality
- $Test_D \in \{\emptyset, 1, 2\}$: Choose no test, Test1, or Test2
- $Test_1, Test_2 \in \{g, b\}$: test yields that the tested car is of good/bad quality
- $Buy_D \in \{1, 2\}$: which car to buy
- $U_{C_1}, U_{C_2} \in Nat$: utility in \$ of buying car 1 resp. car 2

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3. Draw the decision network for which test to apply and which car to buy in either case. This should include:
   - an action node for the test decision (no test, Test1, or Test2)
   - an action node for which car to buy
   - utility nodes for each of the two cars
   - chance nodes for the quality of the cars and the outcomes of the tests

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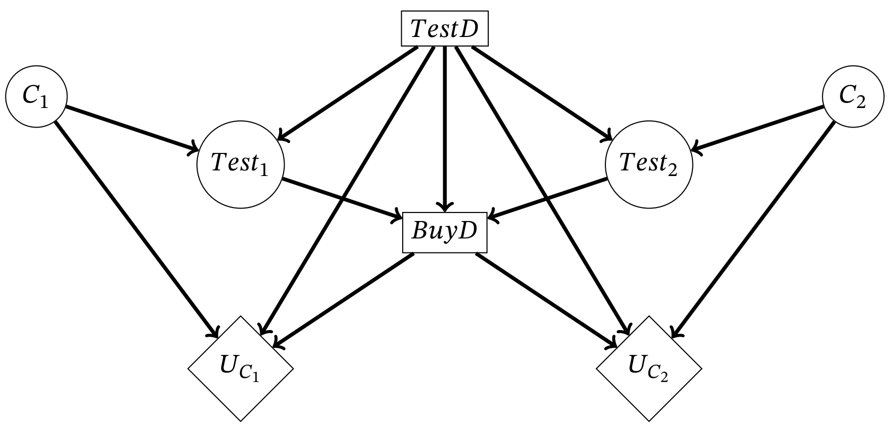
Solution:



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4. How would we choose which test to do?

Solution: For each value of $Test_D$, we calculate the utility of the test by computing the utility for each possible test result.

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5. Assume we have chosen to do $Test_1$ and the outcome was good. Compute which car to buy.

Solution: We have already made one decision: $P(Test_D = 1) = 1$. Our evidence is the event $Test_1 = g$. We need to set $BuyD$ to each of its values and compute the expected utility in each case.
- $BuyD = 1$: We have $P(Test_1 = g \mid C_1 = g) = .85$ and $P(Test_1 = g \mid C_1 =$

$b) = .35$. We compute $\langle P(C_1 = g \mid Test_1 = g), P(C_1 = b \mid Test_1 = g)\rangle$ using

$$P(C_1 \mid Test_1 = g) = \alpha P(C_1, Test_1 = g) = \alpha P(C_1) * P(Test_1 = g \mid C_1)$$

and obtain

$$\alpha\langle .7 * .85, .3 * .35\rangle = \langle .7 * .85, .3 * .35\rangle / .7 = \langle .85, .15\rangle$$

We need to compute $E(U_{C_1}) = P(C_1 = g \mid Test_1 = g) * (2000 - 1500 - 50) + P(C_1 = b \mid Test_1 = g) * (2000 - 1500 - 50 - 700) = 345$

- $BuyD = 2$: $Test_1$ is independent of C_2. So we use $P(C_2 = g) = .65$ and $P(C_2 = b) = .35$. We need to compute $E(U_{C_2}) = P(C_2 = g) * (1400 - 1150 - 50) + P(C_2 = b) * (1400 - 1150 - 50 - 150) = 147.5$.

So we should buy car 1.